THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of steps the Collatz sequence from 6 takes to reach 1 is: 1

(a) 10 **(b)** 7 **(c)** 9 **(d)** 8

2. The smallest digit $*$ making $8*4$ divisible by 9 is: 1

(a) 7 **(b)** 5 **(c)** 8 **(d)** 6

3. The sum $1+2+3+\cdots+20$ is: 1

(a) 212 **(b)** 210 **(c)** 211 **(d)** 209

4. A number is divisible by 2 if its last digit is: 1

(a) 0 or 5 **(b)** 1, 3 or 7 **(c)** 0, 2, 4, 6 or 8 **(d)** any odd digit

5. The sum of five consecutive numbers with middle number 13 is: 1

(a) 64 **(b)** 66 **(c)** 67 **(d)** 65

6. The largest number that can be made using the digits $8, 4, 2, 0$ once each is: 1

(a) 8420 **(b)** 8204 **(c)** 0248 **(d)** 0284

7. A two-digit number has digits 4 and 8. The sum of the number and its reverse is: 1

(a) 134 **(b)** 131 **(c)** 133 **(d)** 132

8. A two-digit number has digits 6 (tens) and 1 (units). Number minus its reverse is: 1

(a) 44 **(b)** 47 **(c)** 46 **(d)** 45

9. The number of steps the Collatz sequence from 7 takes to reach 1 is: 1

(a) 16 **(b)** 18 **(c)** 17 **(d)** 15

10. The smallest digit $*$ making $6*7$ divisible by 9 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the digit $*$ so that $73*5$ is divisible by 9. 2

12. The sum of five consecutive even numbers is 180. Find the numbers. 2

13. Show that $(10a+b)+(10b+a)=11(a+b)$ and verify for $a=2$, $b=9$. 2

14. Find the digit $*$ so that $1*962$ is divisible by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply Kaprekar's routine twice to 8420. 3

16. Find how many reverse-and-add steps turn 59 into a palindrome, showing each step. 3

17. Can 111 be written as the sum of three consecutive numbers? If yes, find them. 3

OR

Apply Kaprekar's routine twice to 6082. 3

18. Find how many reverse-and-add steps turn 38 into a palindrome, showing each step. 3

OR

Can 145 be written as the sum of three consecutive numbers? If yes, find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. For the three-digit number 813, find its reverse, the difference, and show that the difference is divisible by 99. 5

OR

Write the Collatz sequence from 21. Count how many times the number is halved and how many times it is tripled and increased by 1. 5

20. For the three-digit number 651, find its reverse, the difference, and show that the difference is divisible by 99. 5

OR

Write the Collatz sequence from 22. Count how many times the number is halved and how many times it is tripled and increased by 1. 5

END OF QUESTION PAPER — Total: 40 Marks

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